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Find the general solution of the given equation.

$$y'' - 5y' - 6y = 0$$

$y = e^{rx}$ is a solution to the second-order homogeneous differential equation of the form $ay'' + by' + cy = 0$ if and only if r is a solution to $ar^2 + br + c = 0$. The equation $ar^2 + br + c = 0$ is called the auxiliary equation to $ay'' + by' + cy = 0$.

Write the auxiliary equation for the given differential equation.

$$r^2 - 5r - 6 = 0$$

Find the roots of the auxiliary equation to determine the form of the general solution to $y'' - 5y' - 6y = 0$.

$$r^2 - 5r - 6 = 0$$

$$(r - 6)(r + 1) = 0 \quad \text{Factor.}$$

$$r = 6 \text{ or } -1 \quad \text{Solve.}$$

The auxiliary equation has two, distinct real roots. Therefore, the general solution to the given differential equation is of the form shown below.

$$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

Write the general solution to $y'' - 5y' - 6y = 0$ by substituting the solutions to the auxiliary equation $ar^2 + br + c = 0$, $r_1 = 6$ and $r_2 = -1$, into the general solution.

$$y(x) = c_1 e^{6x} + c_2 e^{-x}$$